

Omega-Wide Diameters of Pyramid Networks

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Abstract

This work computes ω -wide diameters of a pyramid network, which is one of the important architectures in parallel computing, network computing, and image processing. The ω -wide diameter of a network, proposed by Hsu in 1994, is very valuable to indicate some important properties of a network, such as reliability, fault-tolerant ability, maximum parallelism and minimum transmission delay of the network. Therefore, to evaluate the fault-tolerant property of a network, determining the wide diameter of the network is a very important issue. In 1999, Cao et al. determined that the 2-wide diameter of an n -layer pyramid network is $3n-1$ and showed that the 3-wide diameter of an n -layer pyramid network is bounded above by $10n/3+6$. This study first investigates the node-disjoint paths between any two distinct nodes and then computes upper and lower bounds of the 3-wide diameter of an n -layer pyramid network. After optimizing upper and lower bounds, the 3-wide diameter of an n -layer pyramid network can then be determined. Our results show that the 3-wide diameter of an n -layer pyramid network is $\lceil (10n+2)/3 \rceil$.

1 Introduction

The performance of an interconnection network can be determined by its topological properties [32]. When an interconnection network is discussed for simulation and communication, many topological properties of the interconnection network are very important, such as embedding, fault tolerance, routing, and wide diameter. In the recent decades, there are many researches on topological properties of interconnection networks,

such as Hamiltonicity [4, 6, 7, 12, 13, 18, 31], pancyclicity [3, 9, 19, 23], fault tolerance [1, 14, 15, 20, 29], wide diameter [27], Rabin numbers [24, 25], routing [8, 16, 30], broadcasting [5, 17, 28], and container [1, 14, 26, 27]. Pyramid networks have conventionally been adopted for image processing [11, 22], computer vision [11], parallel computing [9, 10] and network computing [1]. A pyramid network (PM for short) is a hierarchy structure based on meshes. Note that the node degree of a PM is from 3 to 9, and both node connectivity and edge connectivity are 3 [1, 31].

The topological structure of an interconnection network (network for short) can be modeled by a graph [2, 13, 14, 15, 18, 19, 20, 23, 24, 25, 26, 29, 32]. The vertices and edges of a graph respectively correspond to nodes and links (edges) of an interconnection network. The *length* of a path is the number of its edges, which the path passes through. Given a source node s and a destination node t in a network G , the *distance* between them, denoted by $d(G; s, t)$, is the length of their shortest path. For simplicity, the distance between nodes s and t in a PM is denoted as $d(s, t)$. The *diameter* of a network G , denoted by $d(G)$, is defined as the maximum of $d(G; s, t)$ among all pairs of distinct nodes in G . Obviously, the length of each shortest path between any two distinct nodes in a network G would be bounded above by $d(G)$.

Constructing node-disjoint paths and determining ω -wide diameters of the network are very important issues when estimate reliability, fault-tolerant ability, maximum parallelism and minimum transmission delay of a network. A *container* between two nodes s and t in G , denoted by $\mathcal{C}(s, t)$, is a set of node-disjoint (disjoint for short) paths between s and t [21]. The *width* of $\mathcal{C}(s, t)$ is its cardinality. The ω -wide container, denoted by $\mathcal{C}_\omega(s, t)$, is a $\mathcal{C}(s, t)$ of width ω . Notably, by Menger's theorem, ω is at most the node connectivity of G [20, 32]. The *length* of $\mathcal{C}_\omega(s, t)$,

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denoted by $l(\mathcal{C}_\omega(s, t))$, is the length of the longest path in $\mathcal{C}_\omega(s, t)$. The *total length of $\mathcal{C}_\omega(s, t)$* , denoted by $l_T(\mathcal{C}_\omega(s, t))$, is the sum of the lengths of the ω paths in $\mathcal{C}_\omega(s, t)$. The ω -wide distance between s and t in G , denoted by $d_\omega(G; s, t)$, is the minimum among all $l(\mathcal{C}_\omega(s, t))$ [21]. A best container of width ω between s and t , denoted by $\mathcal{C}_\omega^*(s, t)$, is the $\mathcal{C}_\omega(s, t)$ whose length is $d_\omega(s, t)$. The ω -wide diameter of G , written as $d_\omega(G)$, is the maximum of ω -wide distance among all pairs of distinct nodes in G [21]. Significantly, $d(G; s, t) \leq d_\omega(G; s, t)$, $d(G) \leq d_\omega(G)$, and $d_\omega(G; s, t) \leq d_\omega(G)$. For simplicity, the ω -wide distance between s and t in a PM is denoted as $d_\omega(s, t)$ and the ω -wide diameter of a PM is denoted as $d_\omega(\text{PM})$.

Parallel transmission is a one-to-one communication in a network and the messages can be transmitted from a source node to a destination node via a container between them to enhance the transmission performance and/or improve fault tolerance ability of the communication. Notably, the parallel transmission delay is at most the ω -wide diameter of the network. In 1999, Cao et al. determined that the 2-wide diameter of an n -layer pyramid network is $3n-1$ and showed that 3-wide diameter of an n -layer pyramid network is bounded above by $10n/3+6$ [1]. This work determines the 3-wide diameter of a PM. First, a 3-wide container is built for every two distinct nodes in a PM. Second, an upper bound of $d_3(\text{PM})$ can be determined by computing the maximum length of the constructed containers. Third, a lower bound of $d_3(\text{PM})$ can be obtained by determining the lower bound of the length of a best container between a specific pair of nodes in a PM. Finally, it can be shown that the computed upper and lower bounds of $d_3(\text{PM})$ are the same and thereby $d_3(\text{PM})$ can be determined.

The remainder of this paper is organized as follows. Section 2 describes the structure and terms of a PM, and some notations and definitions in graphs. Section 3 constructs a 3-wide container between every two distinct nodes in a PM and then based on the constructed containers an upper bound of $d_3(\text{PM})$ can be determined. Additionally, a lower bound of $d_3(\text{PM})$ is obtained by computing the lower bound of the 3-wide distance of a specific node pair. Thus, $d_3(\text{PM})$ can be determined. Finally, conclusions are made in Section 4.

2 Structure of a PM

This section formally describes the structure of a PM and then defines some notations and definitions in graphs that are used in the rest of this work.

Let $V(G)$ ($E(G)$) denote the node (edge) set of a network G . The node set of a mesh $M(m, n)$ is $V(M(m, n)) = \{(x, y) \mid 0 \leq x < m, 0 \leq y < n\}$. Two nodes (x_1, y_1) and (x_2, y_2) are joined by an edge (*mesh edge*) if and only if $|x_1 - x_2| + |y_1 - y_2| = 1$, where (x_1, y_1) and (x_2, y_2) belong to $V(M(m, n))$. An n -layer pyramid network, denoted by $\text{PM}[n]$, is a hierarchy structure based on meshes. The node set of $\text{PM}[n]$ is $V(\text{PM}[n]) = \{(l; x, y) \mid 0 \leq l \leq n \text{ and } 0 \leq x, y < 2^l\}$, where $n \geq 1$. A node $(l; x, y) \in V(\text{PM}[n])$ is said to be a node at layer l with coordinate (x, y) in an $M(2^l, 2^l)$ (M_l for short). The nodes at layer l form a mesh, M_l . In general, the node $(0; 0, 0)$ is called the *apex* of $\text{PM}[n]$ (apex for short). For ease of discussion, some symbols are defined in the following.

For a node $v = (l; x, y)$ at layer $1 \leq l \leq n$ with coordinate (x, y) in $\text{PM}[n]$, the coordinate of its *parent*, denoted by $P(v)$, is given by $(l-1; \lfloor x/2 \rfloor, \lfloor y/2 \rfloor)$. Conversely, v is a *child* of $P(v)$. Moreover, each node in $\text{PM}[n]$ has a parent except the apex; and four children except the nodes at layer n . More generally, the k th ancestor of v , denoted by $P^k(v)$, can be recursively defined as follows:

- (1) $k=1$, $P^1(v) = P(v)$ is simply the parent of v .
- (2) $k>1$, $P^k(v) = (l-k; \lfloor x/2^k \rfloor, \lfloor y/2^k \rfloor)$ is the parent of $P^{k-1}(v)$.

A child of v is denoted as $C(v)$, and the coordinates of its four children are given by $(l+1; 2x, 2y)$, $(l+1; 2x+1, 2y)$, $(l+1; 2x, 2y+1)$, and $(l+1; 2x+1, 2y+1)$. Conversely, v is the parent of its children. Also, a k th descendant of v is denoted as $C^k(v)$.

For a node $v = (l; x, y)$ at layer $1 \leq l \leq n$ in $\text{PM}[n]$, $(l; x+1, y)$, $(l; x, y+1)$, $(l; x-1, y)$, and $(l; x, y-1)$ are the coordinates of its four *neighbors* at layer l , and they are also denoted by $N_0(v)$, $N_1(v)$, $N_2(v)$, and $N_3(v)$, respectively. The node v is also a neighbor of its neighbors conversely. Notably, each node v at layer 1 has two neighbors and the apex has no neighbors. There are four *corner nodes* (corners for short) at layer l , $(l; 0, 0)$, $(l; 2^l-1, 0)$, $(l; 0, 2^l-1)$ and $(l; 2^l-1, 2^l-1)$, each of them has two neighbors. Significantly, the apex has no parent, or neighbors, but has four children. In other words, the degree of the apex is 4. Each corner at layer n has no children, but two neighbors and a parent, its degree is 3. Obviously, the node degree of $\text{PM}[n]$ is from 3 to 9. Both the node connectivity and edge connectivity of $\text{PM}[n]$ are 3 [1].

3 Disjoint Paths

Before deriving algorithms for constructing 3 disjoint paths between any pair of nodes in $\text{PM}[n]$, some lemmas are first presented. These lemmas

related to how to route paths in $PM[n]$ are stated in Section 3.1. With the aid of these lemmas, the structure of a shortest path between any pair of nodes is first demonstrated and then the 3 disjoint paths can be built based on the shortest path. Section 3.2 describes algorithms for constructing these 3 disjoint paths. A 3-wide container is built between any two distinct nodes in $PM[n]$. According to the constructed containers, an upper bound of $d_3(PM[n])$ can be determined. Section 3.3 computes $d_3(PM[n])$. First, a lower bound of $d_3(PM[n])$ can be obtained by determining the lower bound of the length of a best container between a specific node pair in $PM[n]$. Then, $d_3(PM[n])$ can be determined because the computed upper and lower bounds of $d_3(PM[n])$ are the same.

3.1 Routing in $PM[n]$

Let $\mathcal{P}(G; u, v)$ ($\mathcal{P}^*(G; u, v)$) denote a path (shortest path) between two nodes u and v in a network G . For simplicity, a path (shortest path) between two nodes u and v in a PM is denoted as $\mathcal{P}(u, v)$ ($\mathcal{P}^*(u, v)$). Let $l(\mathcal{P})$ denote the length of a path $\mathcal{P}$. Let $\mathcal{P}_1 \bullet \mathcal{P}_2$ denote the path of concatenating path $\mathcal{P}_2$ to the tail of path $\mathcal{P}_1$. There are four *border paths* (borders for short) in M_l , $\mathcal{P}^*(M_l; (0, 1), (0, 2^l-2))$, $\mathcal{P}^*(M_l; (2^l-1, 1), (2^l-1, 2^l-2))$, $\mathcal{P}^*(M_l; (1, 0), (2^l-2, 0))$, and $\mathcal{P}^*(M_l; (1, 2^l-1), (2^l-2, 2^l-1))$, where $2 \leq l \leq n$. Clearly, if node u is in a border, then u has three neighbors, and it is not a corner. If the x coordinates (y coordinates) of all nodes of two borders are the same, then these two borders are *parallel* to each other. Some results derived by Cao et al. [1] and Hsieh et al. [16] are described first.

Lemma 1 [1]. The node connectivity of $PM[n]$ is 3.

By Menger's theorem [32] and Lemma 1, there is a 3-wide container between any pair of nodes in $PM[n]$.

Lemma 2 [1]. $d_2(PM[n]) = 3n-1$.

Since $d_2(PM[n])$ has been determined by Cao et al., this work only focuses on $d_3(PM[n])$.

Lemma 3 [16]. If u and v are two nodes at layer l of $PM[n]$, then $d(u, P^i(u)) + d(M_{l-i}; P^i(u), P^i(v)) \leq d(M_l; u, v) + d(v, P^i(v))$, for $1 \leq i < l$.

Lemma 3 shows that $\mathcal{P}(M_l; u, v) \bullet \mathcal{P}(v, P^i(v))$ can be replaced by $\mathcal{P}^*(u, P^i(u)) \bullet \mathcal{P}^*(M_{l-i}; P^i(u), P^i(v))$ without increasing length. With the aid of Lemma 3, eventually obtain the following Corollary 4.

Corollary 4. If u and v are two nodes at layer l of $PM[n]$, then $d(M_l; u, v) < d(u, C^i(u)) + d(M_{l-i}; C^i(u), C^i(v)) + d(C^i(v), v)$, for $l-i \leq n$.

Corollary 4 indicates that path $\mathcal{P}(u, v) = \mathcal{P}^*(u, C^i(u)) \bullet \mathcal{P}^*(M_{l-i}; C^i(u), C^i(v)) \bullet \mathcal{P}^*(C^i(v), v)$ can be replaced by a shorter path $\mathcal{P}^*(M_l; u, v)$ to shorten its length.

Lemma 5 [16]. If $d(M_l; u, v) < 2 + d(M_{l-1}; P(u), P(v))$, then $d(u, v) = d(M_l; u, v) \leq 4$.

According to Lemma 3, Corollary 4 and Lemma 5, we have the following lemma.

Lemma 6 [16]. Given two nodes u and v at layers l and m in $PM[n]$, respectively, where $0 \leq m \leq l \leq n$. Then, there is a $\mathcal{P}^*(u, v)$ with the form $\mathcal{P}^*(u, P^{l-m+i}(u)) \bullet \mathcal{P}^*(M_{m-i}; P^{l-m+i}(u), P^i(v)) \bullet \mathcal{P}^*(P^i(v), v)$, where $0 \leq i \leq m$.

According to Lemma 6, the shape of a shortest path $\mathcal{P}^*(u, v)$ can be described, which is the basis of constructing $\mathcal{C}_3(u, v)$.

For simplicity, let $(a)_b$ denote a modulo b . A path $\mathcal{P}^\#(N_i(u), N_i(P^k(u)))$, where $0 \leq i \leq 3$, disjoint to $\mathcal{P}^*(u, P^k(u))$ can be recursively built as follows:

- (1) $k=1$: If $P(u)=P(N_i(u))$, then $\mathcal{P}^\#(N_i(u), N_i(P(u))) = \mathcal{P}^*(N_i(u), N_i(N_i(u))) \bullet \mathcal{P}^*(N_i(N_i(u)), N_i(P(u)))$, otherwise $\mathcal{P}^\#(N_i(u), N_i(P(u))) = \mathcal{P}^*(N_i(u), N_i(P(u)))$.
- (2) $k>1$: $\mathcal{P}^\#(N_i(u), N_i(P^k(u))) = \mathcal{P}^\#(N_i(u), N_i(P(u))) \bullet \mathcal{P}^\#(N_i(P(u)), N_i(P^k(u)))$.

Significantly, if $P(u)=P(N_i(u))$, then $l(\mathcal{P}^\#(N_i(u), N_i(P(u))))=2$ and $l(\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P(u))))=1$ as shown in Fig. 1(a), otherwise $l(\mathcal{P}^\#(N_i(u), N_i(P(u))))=1$ and $l(\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P(u))))=2$ as shown in Fig. 1(b). Therefore, $l(\mathcal{P}^\#(N_i(u), N_i(P(u)))) + l(\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P(u)))) = 3$. Fig. 1 represents the relation of three disjoint paths $\mathcal{P}^*(u, P(u))$, $\mathcal{P}^\#(N_i(u), N_i(P(u)))$ and $\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P(u)))$. According to Fig. 1, $\mathcal{P}^\#(N_i(u), N_i(P^k(u)))$ can be easily derived. Naturally, $\mathcal{P}^\#(N_i(u), N_i(P^k(u)))$ and $\mathcal{P}^\#(N_i(P^k(u)), N_i(u))$ are the same.

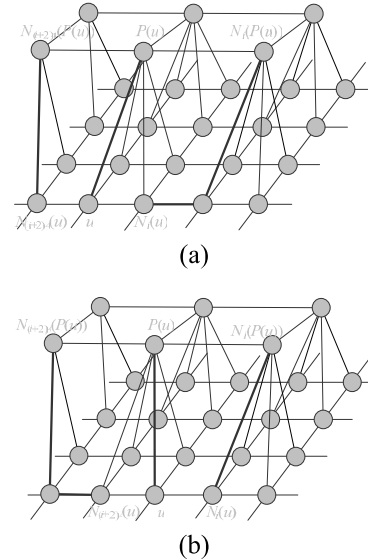


Fig. 1. $\mathcal{P}^*(u, P(u))$, $\mathcal{P}^\#(N_i(u), N_i(P(u)))$ and $\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P(u)))$. (a) $P(u) = P(N_i(u))$. (b) $P(u) \neq P(N_i(u))$.

Lemma 7. $l(\mathcal{P}^\#(N_i(u), N_i(P^k(u)))) + l(\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P^k(u)))) = 3k$.

Proof. Based on the structure of $PM[n]$, referring to Fig. 1, $l(\mathcal{P}^\#(N_i(u), N_i(P(u))) + l(\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P(u)))) = 3$. Consequently, $l(\mathcal{P}^\#(N_i(u), N_i(P^k(u)))) + l(\mathcal{P}^\#(N_{(i+2)_4}(u), N_{(i+2)_4}(P^k(u)))) = 3k$. ■

3.2 ω -Wide Containers

This section constructs a ω -wide container between any pair of nodes in $PM[n]$. For ease of discussion, let $s = (l; x_s, y_s)$, $t = (m; x_t, y_t)$ be any two distinct nodes in $PM[n]$ for $n \geq 4$, where $2 \leq m \leq l \leq n$. Obviously, $d(M_m; P^{l-m}(s), t) = \lfloor x_t - x_s / 2^{l-m} \rfloor + \lfloor y_t - y_s / 2^{l-m} \rfloor$. Without loss of generality, assume $\lfloor y_t - y_s / 2^{l-m} \rfloor \leq \lfloor x_t - x_s / 2^{l-m} \rfloor$.

Let $\mathcal{C}_3(s, t) = \{\mathcal{P}_1(s, t), \mathcal{P}_2(s, t), \mathcal{P}_3(s, t)\}$. Before forming $\mathcal{C}_3(s, t)$, $\mathcal{P}_1(s, t) = \mathcal{P}^*(s, s') \bullet \mathcal{P}(s', t') \bullet \mathcal{P}^*(t', t)$, $\mathcal{P}_2(s, t) = \mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(s')) \bullet \mathcal{P}(N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$ and $\mathcal{P}_3(s, t) = \mathcal{P}^*(s, N_g(s)) \bullet \mathcal{P}^\#(N_g(s), N_g(s')) \bullet \mathcal{P}(N_g(s'), N_h(t')) \bullet \mathcal{P}^\#(N_h(t'), N_h(t)) \bullet \mathcal{P}^*(N_h(t), t)$ can be built by Algorithm Build_3_ $\mathcal{P}(s, t)$ as follows. Let $\ell = \max\{\min\{l \mid v = (l; x, y) \in V(\mathcal{P}_1(s, t))\}, \min\{l \mid v = (l; x, y) \in V(\mathcal{P}_2(s, t))\}, \min\{l \mid v = (l; x, y) \in V(\mathcal{P}_3(s, t))\}\}$ such that $s' = P^{l-\ell}(s)$ and $t' = P^{m-\ell}(t)$ which are two regarded nodes used in Algorithm Build_3_ $\mathcal{P}(s, t)$ for building these three paths mentioned above.

Algorithm Build_3_ $\mathcal{P}(s, t)$. // Construct $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ based on a $\mathcal{P}^*(s, t)$ built according to Lemma 6.
// Output: $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$.
{

$\mathcal{P}^*(s, t) = \mathcal{P}^*(u, P^{l-m+i}(u)) \bullet \mathcal{P}^*(M_{m-i}, P^{l-m+i}(u), P^i(v)) \bullet \mathcal{P}^*(P^i(v), v)$ is built via Lemma 6.

Step 1: // Determine ℓ for constructing paths.

$\ell \leftarrow m-i$; // ℓ is initialized to the top layer of the $\mathcal{P}^*(s, t)$.

While $(\lfloor x_t / 2^{m-\ell} \rfloor - \lfloor x_s / 2^{l-\ell} \rfloor) < 2$ and $(\lfloor y_t / 2^{m-\ell} \rfloor - \lfloor y_s / 2^{l-\ell} \rfloor) < 2$

$\ell \leftarrow \ell + 1$;

EndWhile

// After Step 1, $\ell \geq 2$, $s' = P^{l-\ell}(s)$ and $t' = P^{m-\ell}(t)$.

Step 2: // Build $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$ and $\mathcal{P}(N_g(s'), N_h(t'))$.

Case 1. (s' or t' is a corner):

Select i, j, g , and h , such that they satisfy the following three conditions.

1. If s' is a corner, then one neighbor of s' is $N_i(s')$, the other one is $N_g(s')$. Otherwise, $N_g(s')$ is $N_{(i+2)_4}(s')$. // s' has two neighbors.
2. If t' is a corner, then one neighbor of t' is $N_j(t')$, the other one is $N_h(t')$. Otherwise, $N_h(t')$ is $N_{(j+2)_4}(t')$. // t' has two neighbors.

3. A $\mathcal{P}^*(M_g; N_i(s'), N_j(t'))$ is disjoint to $\mathcal{P}^*(M_g; N_g(s'), N_h(t'))$.
 $\mathcal{P}(s', t') \leftarrow \mathcal{P}^*(s', P(s')) \bullet \mathcal{P}^*(P(s'), P(t')) \bullet \mathcal{P}^*(P(t'), t')$;
 $\mathcal{P}(N_i(s'), N_j(t')) \leftarrow \mathcal{P}^*(M_g; N_i(s'), N_j(t'))$;
 $\mathcal{P}(N_g(s'), N_h(t')) \leftarrow \mathcal{P}^*(M_g; N_g(s'), N_h(t'))$;
// Refer to Fig. 2 (a)

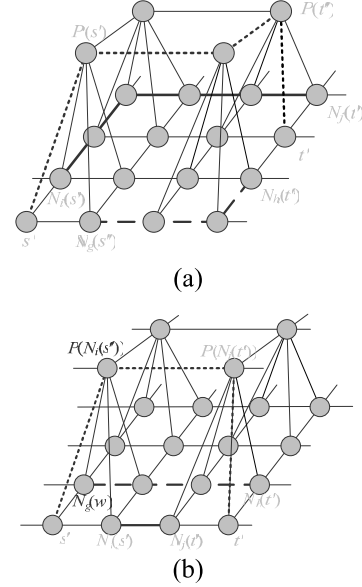


Fig. 2. The constructions of $\mathcal{P}(s', t')$ (dotted line), $\mathcal{P}(N_i(s'), N_j(t'))$ (solid line) and $\mathcal{P}(N_g(s'), N_h(t'))$ (dashed line). (a) s' is a corner but t' is not a corner. (b) Both s' and t' are in a same border.

Case 2. (s' and t' are in parallel borders):

Select i , such that both $N_i(s')$ and $N_{(i+2)_4}(s')$ exist.

$j = i, h = g = (i+2)_4$;

$\mathcal{P}(s', t') \leftarrow \mathcal{P}^*(M_g; s', t')$;

$\mathcal{P}(N_i(s'), N_j(t')) \leftarrow \mathcal{P}^*(N_i(s'), N_j(t'))$;

$\mathcal{P}(N_g(s'), N_h(t')) \leftarrow \mathcal{P}^*(N_g(s'), N_h(t'))$;

Case 3. (s' and t' are in a same border):

Let $N_i(s')$ and $N_j(t')$ be nodes in the border, where $j = (i+2)_4$, such that $\mathcal{P}^\#(N_i(s'), N_i(s')) \bullet \mathcal{P}^*(M_g; N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t'))$ is not longer than $\mathcal{P}^\#(N_{(i+2)_4}(s'), N_{(i+2)_4}(s')) \bullet \mathcal{P}^*(M_g; N_{(i+2)_4}(s'), N_{(j+2)_4}(t')) \bullet \mathcal{P}^\#(N_{(j+2)_4}(t'), N_{(j+2)_4}(t'))$. Also let $N_g(s')$ ($N_h(t')$) be the neighbor of s' (t') and not on the border.

$\mathcal{P}(N_g(s'), N_h(t')) \leftarrow \mathcal{P}^*(M_g; N_g(s'), N_h(t'))$;

Case 3.1. (Both $N_i(s')$ and $N_j(t')$ are in $\mathcal{P}^*(M_g; s', t')$):

$\mathcal{P}(s', t') \leftarrow \mathcal{P}^*(s', P(s')) \bullet \mathcal{P}^*(P(s'), P(t')) \bullet \mathcal{P}^*(P(t'), t')$;

$\mathcal{P}(N_i(s'), N_j(t')) \leftarrow \mathcal{P}^*(M_g; N_i(s'), N_j(t'))$;

// Refer to Fig. 2(b)

Case 3.2. (Otherwise):

$\mathcal{P}(s', t') \leftarrow \mathcal{P}^*(M_\epsilon; s', t')$;
 $\mathcal{P}(N_i(s'), N_j(t')) \leftarrow \mathcal{P}^*(N_i(s'), P(N_i(s')))$
 $\bullet \mathcal{P}^*(P(N_i(s')), P(N_j(t')) \bullet \mathcal{P}^*(P(N_j(t')),$
 $N_j(t'))$;
 Case 4. (Otherwise):

Select i, j, g , and h , such that they satisfy the following two conditions.

1. One of $d(M_\epsilon; N_i(s'), N_j(t'))$ and $d(M_\epsilon; N_g(s'), N_h(t'))$ is $d(M_\epsilon; s', t') + 2$, the other one is $d(M_\epsilon; s', t') - 2$.
2. $\mathcal{P}^\#(N_i(s), N_i(s')) \bullet \mathcal{P}^*(M_\epsilon; N_i(s'), N_j(t'))$
 $\bullet \mathcal{P}^\#(N_j(t'), N_j(t))$ is not longer than
 $\mathcal{P}^\#(N_g(s), N_g(s')) \bullet \mathcal{P}^*(M_\epsilon; N_g(s'), N_h(t'))$
 $\bullet \mathcal{P}^\#(N_h(t'), N_h(t))$, where $N_g(s')$ is $N_{(i+2)_4}(s')$ and $N_h(t')$ is $N_{(j+2)_4}(t')$.

$\mathcal{P}(s', t') \leftarrow \mathcal{P}^*(M_\epsilon; s', t')$;

Case 4.1 (Both $N_i(s')$ and $N_j(t')$ are in a $\mathcal{P}^*(M_\epsilon; s', t')$):

$\mathcal{P}(N_i(s'), N_j(t')) \leftarrow \mathcal{P}^*(M_\epsilon; N_i(s'), N_j(t'))$;
 $\mathcal{P}(N_g(s'), N_h(t')) \leftarrow \mathcal{P}^*(N_g(s'), P(N_g(s')))$
 $\bullet \mathcal{P}^*(P(N_g(s')), P(N_h(t')) \bullet$
 $\mathcal{P}^*(P(N_h(t')), N_h(t'))$;

Case 4.2 (Otherwise):

$\mathcal{P}(N_i(s'), N_j(t')) \leftarrow \mathcal{P}^*(N_i(s'), P(N_i(s')) \bullet$
 $\mathcal{P}^*(P(N_i(s')), P(N_j(t')) \bullet \mathcal{P}^*(P(N_j(t')),$
 $N_j(t'))$;
 $\mathcal{P}(N_g(s'), N_h(t')) \leftarrow \mathcal{P}^*(M_\epsilon; N_g(s'),$
 $N_h(t'))$;

Step 3: // Build paths $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ such that $l(\mathcal{P}_2(s, t)) \leq l(\mathcal{P}_3(s, t))$.

$\mathcal{P}_1(s, t) \leftarrow \mathcal{P}^*(s, s') \bullet \mathcal{P}(s', t') \bullet \mathcal{P}^*(t', t)$;
 $\mathcal{P}_2(s, t) \leftarrow \mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(s')) \bullet$
 $\mathcal{P}(N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t)) \bullet \mathcal{P}^*(N_j(t),$
 $t)$;
 $\mathcal{P}_3(s, t) \leftarrow \mathcal{P}^*(s, N_g(s)) \bullet \mathcal{P}^\#(N_g(s), N_g(s'))$
 $\bullet \mathcal{P}(N_g(s'), N_h(t')) \bullet \mathcal{P}^\#(N_h(t'), N_h(t))$
 $\bullet \mathcal{P}^*(N_h(t), t)$;

If $(l(\mathcal{P}_2(s, t)) > l(\mathcal{P}_3(s, t)))$, then exchange $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$. // Let $l(\mathcal{P}_2(s, t)) \leq l(\mathcal{P}_3(s, t))$.

} // End of Algorithm Build_3_ $\mathcal{P}(s, t)$.

For ease of discussion, without loss of generality, assume that $\mathcal{P}_2(s, t)$ is $\mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(s')) \bullet \mathcal{P}(N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$; $\mathcal{P}_3(s, t)$ is $\mathcal{P}^*(s, N_g(s)) \bullet \mathcal{P}^\#(N_g(s), N_g(s')) \bullet \mathcal{P}(N_g(s'), N_h(t')) \bullet \mathcal{P}^\#(N_h(t'), N_h(t)) \bullet \mathcal{P}^*(N_h(t), t)$, and $l(\mathcal{P}_2(s, t)) \leq l(\mathcal{P}_3(s, t))$.

Lemma 8. $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ constructed by Algorithm Build_3_ $\mathcal{P}(s, t)$ form a $\mathcal{C}_3(s, t)$.

Proof. As mentioned above, $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ are composed of $\mathcal{P}^*(s, s')$, $\mathcal{P}(s', t')$ and $\mathcal{P}^*(t', t)$, $\mathcal{P}^*(s, N_i(s))$, $\mathcal{P}^\#(N_i(s), N_i(s'))$, $\mathcal{P}(N_i(s'),$

$N_j(t'))$, $\mathcal{P}^\#(N_j(t'), N_j(t))$ and $\mathcal{P}^*(N_j(t), t)$, and $\mathcal{P}^*(s, N_g(s))$, $\mathcal{P}^\#(N_g(s), N_g(s'))$, $\mathcal{P}(N_g(s'), N_h(t'))$, $\mathcal{P}^\#(N_h(t'), N_h(t))$ and $\mathcal{P}^*(N_h(t), t)$, respectively. According to structures of $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$, $\mathcal{P}^*(s, s')$, $\mathcal{P}^*(t', t)$, $\mathcal{P}^*(s, N_i(s))$, $\mathcal{P}^\#(N_i(s), N_i(s'))$, $\mathcal{P}^\#(N_j(t'), N_j(t))$ and $\mathcal{P}^*(N_j(t), t)$, $\mathcal{P}^*(s, N_g(s))$, $\mathcal{P}^\#(N_g(s), N_g(s'))$, $\mathcal{P}^\#(N_h(t'), N_h(t))$ and $\mathcal{P}^*(N_h(t), t)$ are naturally disjoint except for nodes s and t . Furthermore, $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$, and $\mathcal{P}(N_g(s'), N_h(t'))$ are also disjoint to all nodes at layer greater than ℓ . Therefore, to prove $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ are disjoint, only need to show that $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$ and $\mathcal{P}(N_g(s'), N_h(t'))$ built by Step 2 of Algorithm Build_3_ $\mathcal{P}(s, t)$ are disjoint.

Only some cases are described here, other cases can be proved similarly. Fig. 2(a) shows when $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$ and $\mathcal{P}(N_g(s'), N_h(t'))$ are built by Case 1 and s' is a corner but t' is not a corner, the selected i, j, g , and h satisfy the three conditions listed in Case 1 of Step 2 of Algorithm Build_3_ $\mathcal{P}(s, t)$. Paths $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$ and $\mathcal{P}(N_g(s'), N_h(t'))$ built by Case 3.1 of Step 2 of Algorithm Build_3_ $\mathcal{P}(s, t)$ is shown in Fig. 2(b). Obviously, $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$ and $\mathcal{P}(N_g(s'), N_h(t'))$ are disjoint. Other cases can be discussed similarly, and the built $\mathcal{P}(s', t')$, $\mathcal{P}(N_i(s'), N_j(t'))$ and $\mathcal{P}(N_g(s'), N_h(t'))$ are also disjoint.

Therefore, $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ constructed by Algorithm Build_3_ $\mathcal{P}(s, t)$ are disjoint, and form a $\mathcal{C}_3(s, t)$. ■

In the following, Lemma 9 demonstrates that after running Algorithm Build_3_ $\mathcal{P}(s, t)$ the upper bound of $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t))$ can be computed. Based on the result, $\mathcal{C}_3(s, t)$ can thus be built by Algorithm Build_ $\mathcal{C}_3(s, t)$.

Lemma 9. $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 10n - 4$.

Proof. Clearly, $l(\mathcal{P}^\#(N_i(s), N_i(s')))$, $l(\mathcal{P}^\#(N_j(t'), N_j(t)))$, $l(\mathcal{P}^\#(N_g(s), N_g(s')))$, and $l(\mathcal{P}^\#(N_h(t'), N_h(t)))$ are at most $2(n - \ell)$. Furthermore, according to the structures of paths built by Algorithm Build_3_ $\mathcal{P}(s, t)$ and with the aid of Lemma 7, when $g = (i+2)_4$, $l(\mathcal{P}^\#(N_i(s), N_i(s')) + l(\mathcal{P}^\#(N_g(s), N_g(s')))) \leq 3(l - \ell)$; when $h = (j+2)_4$, $l(\mathcal{P}^\#(N_j(t'), N_j(t)) + l(\mathcal{P}^\#(N_h(t'), N_h(t)))) \leq 3(m - \ell)$. When $\mathcal{P}_1(s, t)$ is built by Cases 1 and 3.1 of Step 2 of Algorithm Build_3_ $\mathcal{P}(s, t)$, according to its structure, $l(\mathcal{P}_1(s, t)) \leq 2n$. Otherwise, $l(\mathcal{P}_1(s, t)) \leq (l - \ell) + d(M_\epsilon; s', t') + (m - \ell)$. Since $l(\mathcal{P}_2(s, t)) \leq l(\mathcal{P}_3(s, t))$, $l(\mathcal{P}_2(s, t)) \leq \lfloor (l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t))) / 2 \rfloor$. In the following, lengths of paths are determined according to cases in Step 2 of

Algorithm Build_3_℘(s, t) as following.

Case 1. (At least one of s' or t' is a corner): $d(M_\delta; s', t') \leq 2 \times (2^\ell - 1)$, $\ell \geq \lg(d(M_\delta; s', t') + 2) - 1$.

Case 1.1 (Both s' and t' are corners): According to the structure of PM[n] and paths built by Algorithm Build_3_℘(s, t), $\ell = 2$ and $l(\mathcal{P}(N_i(s'), N_j(t'))) = l(\mathcal{P}(N_g(s'), N_h(t'))) = 4$. Thus, $l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2 \times (1 + 2(\ell - 2) + 4 + 2(m - 2) + 1) = 4\ell + 4m - 4 \leq 8n - 4$. Hence, $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2n + (8n - 4) = 10n - 4$.

Case 1.2 (One of s' and t' is a corner): $d(M_\delta; s', t') \leq 2 \times (2^\ell - 1) - 1$, $\ell \geq \lg(d(M_\delta; s', t') + 3) - 1$. According to the structures of $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$, $l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2 + 4(n - \ell) + 2d(M_\delta; s', t') + 3(n - \ell) + 2$. After substitution all possible $d(M_\delta; s', t')$ and its relative ℓ , $l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 7n - 5$. Hence, $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2n + (7n - 5) = 9n - 5$.

Case 2 (s' and t' are in two parallel borders): According to the structure of PM[n] and paths built by Algorithm Build_3_℘(s, t), $\ell = 2$, $d(M_\delta; s', t') \leq 4$, and $l(\mathcal{P}(N_i(s'), N_j(t'))) = l(\mathcal{P}(N_g(s'), N_h(t'))) = 3$. Thus, $l(\mathcal{P}_1(s, t)) \leq (\ell - 2) + 4 + (m - 2) \leq 2n$; and $l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2 + 3(\ell - 2) + 3 + 3 + 3(m - 2) + 2 \leq 6n - 2$. That is, $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2n + (6n - 2) = 8n - 2$.

Case 3 (s' and t' are in a same border): $d(M_\delta; s', t') \leq (2^\ell - 1) - 2$, $\ell \geq \lg(d(M_\delta; s', t') + 3)$. According to the structure of $\mathcal{P}_3(s, t)$, $l(\mathcal{P}_3(s, t)) \leq 1 + 2(\ell - \ell) + d(M_\delta; s', t') + 2(m - \ell) + 1 \leq 4(n - \ell) + d(M_\delta; s', t') + 2 \leq 4n - 5$. Since $\mathcal{P}^\#(N_i(s), N_i(s')) \bullet \mathcal{P}^*(T_\delta; N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t))$ is not longer than $\mathcal{P}^\#(N_{(i+2)_4}(s), N_{(i+2)_4}(s')) \bullet \mathcal{P}^*(T_\delta; N_{(i+2)_4}(s'), N_{(j+2)_4}(t')) \bullet \mathcal{P}^\#(N_{(j+2)_4}(t'), N_{(j+2)_4}(t))$, $l(\mathcal{P}^\#(N_i(s), N_i(s')) \bullet \mathcal{P}^*(T_\delta; N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t))) \leq (3(\ell - \ell) + d(M_\delta; s', t') + 2 + d(M_\delta; s', t') - 2 + 3(m - \ell))/2 \leq 3(n - \ell) + d(M_\delta; s', t')$.

Case 3.1 (Both $N_i(s')$ and $N_j(t')$ are in $\mathcal{P}^*(M_\delta; s', t')$): $l(\mathcal{P}_2(s, t)) = l(\mathcal{P}^*(s, N_i(s'))) + l(\mathcal{P}^\#(N_i(s), N_i(s')) \bullet \mathcal{P}^*(M_\delta; N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t))) + l(\mathcal{P}^*(N_j(t), t)) \leq 3(n - \ell) + d(M_\delta; s', t') + 2 \leq 3n - 2$.

Case 3.2 (Otherwise): $l(\mathcal{P}_1(s, t)) \leq 2(n - \ell) + d(M_\delta; s', t') \leq 2n - 1$. The built $\mathcal{P}(N_i(s'), N_j(t'))$ is $\mathcal{P}^*(N_i(s'), P(N_i(s'))) \bullet \mathcal{P}^*(M_{\ell-1}; P(N_i(s')), P(N_j(t')) \bullet \mathcal{P}^*(P(N_j(t')), N_j(t'))$ and $d(T_\delta; N_i(s'), N_j(t')) = d(M_\delta; s', t') + 2$. Thus, $l(\mathcal{P}(N_i(s'), N_j(t'))) \leq \lfloor (d(M_\delta; s', t') + 1 + 2)/2 \rfloor + 2 = \lfloor (d(M_\delta; s', t') + 1)/2 \rfloor + 3$ and $l(\mathcal{P}_2(s, t)) \leq (3(n - \ell) + d(M_\delta; s', t') - (d(M_\delta; s', t') + 2) + \lfloor (d(M_\delta; s', t') + 1)/2 \rfloor + 3) + 2 \leq 3n - 3$.

Hence, $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 9n - 7$.

Case 4 (Otherwise): $\ell > (\lg(d(M_\delta; s', t') + 4)) - 1$. $l(\mathcal{P}_1(s, t)) \leq (\ell - \ell) + d(M_\delta; s', t') + (m - \ell) \leq 2n$, $l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 2 + 3(\ell - \ell) + (d(M_\delta; s', t') - 2) + \lfloor (d(M_\delta; s', t') + 2 + 2)/2 \rfloor + 2 + 3(m - \ell) + 2 = 3(\ell - \ell) + 3(m - \ell) + d(M_\delta; s', t') + \lfloor d(M_\delta; s', t')/2 \rfloor + 6 \leq 6n - 1$. Hence, $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 8n - 1$.

According to the description above, $l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)) \leq 10n - 4$. ■

Naturally, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ constructed by Algorithm Build_3_℘(s, t) may be much longer than $\mathcal{P}_1(s, t)$. $\mathcal{C}_3(s, t)$ can be formed by reconstructing $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ built by Algorithm Build_3_℘(s, t). Obviously, $l(\mathcal{C}_3(s, t))$ is at least $\lceil (l(\mathcal{P}_1(s, t)) + l(\mathcal{P}_2(s, t)) + l(\mathcal{P}_3(s, t)))/3 \rceil$. Therefore, according to Lemma 9, the upper bound of $l(\mathcal{C}_3(s, t))$ is at least $\lceil (10n - 4)/3 \rceil$. Although $\lceil (10n - 4)/3 \rceil$ maybe not the real upper bound of $l(\mathcal{C}_3(s, t))$, it is used to determine the form of $\mathcal{C}_3(s, t)$ for minimizing the upper bound $l(\mathcal{C}_3(s, t))$.

$\mathcal{C}_3(s, t)$ can be formed in three cases. In Case 1, $l(\mathcal{P}_2(s, t)) > \lceil (10n - 4)/3 \rceil$, $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ can be reconstructed based on $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$ built by Algorithm Build_℘(s, t). The reconstructed paths are shown in Fig. 3(a). In Case 2, $l(\mathcal{P}_2(s, t)) \leq \lceil (10n - 4)/3 \rceil < l(\mathcal{P}_3(s, t))$, let $\mathcal{P}_2^3(s, t) = \mathcal{P}_2(s, t)$, $\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ can be reconstructed based on $\mathcal{P}_1(s, t)$ and $\mathcal{P}_3(s, t)$. The reconstructed paths are shown in Fig. 3(b). In Case 3, $l(\mathcal{P}_3(s, t)) \leq \lceil (10n - 4)/3 \rceil$, let $\mathcal{P}_1^3(s, t) = \mathcal{P}_1(s, t)$, $\mathcal{P}_2^3(s, t) = \mathcal{P}_2(s, t)$ and $\mathcal{P}_3^3(s, t) = \mathcal{P}_3(s, t)$, according to Lemma 8, these three paths form a $\mathcal{C}_3(s, t)$, and $l(\mathcal{C}_3(s, t)) \leq \lceil (10n - 4)/3 \rceil$.

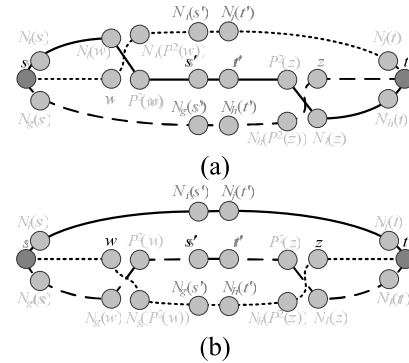


Fig. 3. The construction of $\mathcal{C}_3(s, t)$. The dotted line is $\mathcal{P}_1^3(s, t)$, the solid line is $\mathcal{P}_2^3(s, t)$, and the dashed line is $\mathcal{P}_3^3(s, t)$. (a) $\lceil (10n - 4)/3 \rceil < l(\mathcal{P}_2(s, t))$. (b) $l(\mathcal{P}_2(s, t)) \leq \lceil (10n - 4)/3 \rceil < l(\mathcal{P}_3(s, t))$.

Let $w = P^{\ell - l_w}(s)$, $P^2(w) = P(N_g(P(w)))$ and there are $\lfloor (2 \times (l(\mathcal{P}^\#(N_i(s), N_i(s'))) - l(\mathcal{P}^*(s, s')) - 1)/3 \rfloor - 2$ mesh edges in $\mathcal{P}^\#(N_i(s), N_i(P^2(w)))$; and $z = P^{m - l_z}(t)$, and $P^2(z) = P(N_h(P(z)))$ and there are $\lfloor (2 \times (l(\mathcal{P}^\#(N_h(t'), N_h(t))) - l(\mathcal{P}^*(t', t)) - 1)/3 \rfloor$ mesh

edges in $\mathcal{P}^\#(N_h(z), N_h(P^2(t)))$. In Case 1, $l(\mathcal{P}_2(s, t)) > \lceil (10n-4)/3 \rceil$, after applying Algorithm Build $\mathcal{C}_3(s, t)$, $\mathcal{P}_1^3(s, t) = \mathcal{P}^*(s, w) \bullet \mathcal{P}(w, N_i(P^2(w))) \bullet \mathcal{P}^\#(N_i(P^2(w)), N_i(s')) \bullet \mathcal{P}(N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$, $\mathcal{P}_2^3(s, t) = \mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(w)) \bullet \mathcal{P}(N_i(w), P^2(w)) \bullet \mathcal{P}^*(P^2(w), s') \bullet \mathcal{P}(s', t') \bullet \mathcal{P}^*(t', P^2(z)) \bullet \mathcal{P}(P^2(z), N_h(z)) \bullet \mathcal{P}^\#(N_h(z), N_h(t)) \bullet \mathcal{P}^*(N_h(t), t)$ and $\mathcal{P}_3^3(s, t) = \mathcal{P}^*(s, N_g(s)) \bullet \mathcal{P}^\#(N_g(s), N_g(s')) \bullet \mathcal{P}(N_g(s'), N_h(t')) \bullet \mathcal{P}^\#(N_h(t'), N_h(P^2(z))) \bullet \mathcal{P}(N_h(P^2(z)), z) \bullet \mathcal{P}^*(z, t)$ as shown in (a). Naturally, the built $\mathcal{P}(w, N_i(P^2(w)))$, $\mathcal{P}(N_i(w), P^2(w))$, $\mathcal{P}(P^2(z), N_h(z))$ and $\mathcal{P}(N_h(P^2(z)), z)$ should be disjoint to each other, so that $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ are possibly disjoint to each other.

In Case 2, $l(\mathcal{P}_2(s, t)) \leq \lceil (10n-4)/3 \rceil < l(\mathcal{P}_3(s, t))$, $\mathcal{P}_1^3(s, t) = \mathcal{P}^*(s, w) \bullet \mathcal{P}(w, N_g(P^2(w))) \bullet \mathcal{P}^\#(N_g(P^2(w)), N_g(s')) \bullet \mathcal{P}(N_g(s'), N_h(t')) \bullet \mathcal{P}^\#(N_h(t'), N_h(P^2(z))) \bullet \mathcal{P}(N_h(P^2(z)), z) \bullet \mathcal{P}^*(z, t)$ and $\mathcal{P}_3^3(s, t) = \mathcal{P}^*(s, N_g(s)) \bullet \mathcal{P}^\#(N_g(s), N_g(w)) \bullet \mathcal{P}(N_g(w), P^2(w)) \bullet \mathcal{P}^*(P^2(w), s') \bullet \mathcal{P}(s', t') \bullet \mathcal{P}^*(t', P^2(z)) \bullet \mathcal{P}(P^2(z), N_h(z)) \bullet \mathcal{P}^\#(N_h(z), N_h(t)) \bullet \mathcal{P}^*(N_h(t), t)$ as shown in Fig. 3(b). Naturally, the built $\mathcal{P}(w, N_g(P^2(w)))$, $\mathcal{P}(N_g(s'), N_h(t'))$, $\mathcal{P}(N_g(w), P^2(w))$ and $\mathcal{P}(P^2(z), N_h(z))$ should be disjoint to each other, so that $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ are possibly disjoint to each other.

With the determined w and z , a $\mathcal{C}_3(s, t)$ can be built by Algorithm Build $\mathcal{C}_3(s, t)$. Algorithm Build $\mathcal{C}_3(s, t)$ is formally stated in the following.

Algorithm Build $\mathcal{C}_3(s, t)$ // Construct $\mathcal{C}_3(s, t)$ based on $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$ and $\mathcal{P}_3(s, t)$.

// Input: $\mathcal{P}_1(s, t)$, $\mathcal{P}_2(s, t)$, $\mathcal{P}_3(s, t)$, w and z .

// Output: A $\mathcal{C}_3(s, t) = \{\mathcal{P}_1^3(s, t), \mathcal{P}_2^3(s, t), \mathcal{P}_3^3(s, t)\}$.

{
 Step 1. // Build $\mathcal{P}(w, N_i(P^2(w)))$, $\mathcal{P}(N_i(w), P^2(w))$, $\mathcal{P}(N_h(P^2(z)), z)$, and $\mathcal{P}(P^2(z), N_h(z))$.
 $\mathcal{P}(w, N_i(P^2(w))) \leftarrow \mathcal{P}^*(w, P(w)) \bullet \mathcal{P}^*(P(w), N_i(P(w))) \bullet \mathcal{P}^\#(N_i(P(w)), N_i(P^2(w)))$;
 $\mathcal{P}(N_i(w), P^2(w)) \leftarrow \mathcal{P}^*(T_{lw}; N_i(w), a) \bullet \mathcal{P}^*(a, P^2(w))$;
 // Referring to Fig. 4, a is a $C^2(P^2(w))$, $l(\mathcal{P}(w, N_i(P^2(w)))) = 4$ and $l(\mathcal{P}(N_i(w), P^2(w))) = 4$ (5) when $P(w) \neq P(N_i(w))$ ($P(w) = P(N_i(w))$).
 $\mathcal{P}(N_h(P^2(z)), z) \leftarrow \mathcal{P}^\#(N_h(P^2(z)), N_h(P(z))) \bullet \mathcal{P}^*(N_h(P(z)), P(z)) \bullet \mathcal{P}^*(P(z), z)$;
 $\mathcal{P}(P^2(z), N_h(z)) \leftarrow \mathcal{P}^*(P^2(z), b) \bullet \mathcal{P}^*(T_{lz}; b, N_h(z))$;
 // $l(\mathcal{P}(N_h(P^2(z)), z)) = 4$ and $l(\mathcal{P}(P^2(z), N_h(z))) = 4$ (5) when $P(z) \neq P(N_h(z))$ ($P(z) = P(N_h(z))$).
 Step 2. // Build $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$.
 $\mathcal{P}_1^3(s, t) \leftarrow \mathcal{P}^*(s, w) \bullet \mathcal{P}(w, N_i(P^2(w))) \bullet \mathcal{P}^\#(N_i(P^2(w)), N_i(s')) \bullet \mathcal{P}(N_i(s'), N_j(t')) \bullet \mathcal{P}^\#(N_j(t'), N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$;

$\mathcal{P}_2^3(s, t) \leftarrow \mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(w)) \bullet \mathcal{P}(N_i(w), P^2(w)) \bullet \mathcal{P}^*(P^2(w), s') \bullet \mathcal{P}(s', t') \bullet \mathcal{P}^*(t', P^2(z)) \bullet \mathcal{P}(P^2(z), N_h(z)) \bullet \mathcal{P}^\#(N_h(z), N_h(t)) \bullet \mathcal{P}^*(N_h(t), t)$;
 $\mathcal{P}_3^3(s, t) \leftarrow \mathcal{P}^*(s, N_g(s)) \bullet \mathcal{P}^\#(N_g(s), N_g(s')) \bullet \mathcal{P}(N_g(s'), N_h(t')) \bullet \mathcal{P}^\#(N_h(t'), N_h(P^2(z))) \bullet \mathcal{P}(N_h(P^2(z)), z) \bullet \mathcal{P}^*(z, t)$;
 } // End of Algorithm Build $\mathcal{C}_3(s, t)$.

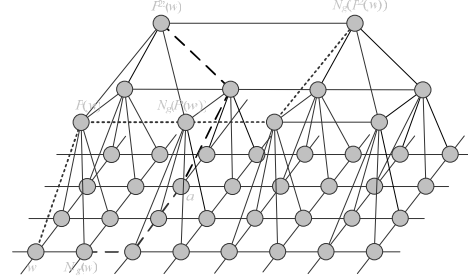


Fig. 4. The constructions of $\mathcal{P}(w, N_i(P^2(w)))$ and $\mathcal{P}(N_i(w), P^2(w))$.

According to the structures of $\mathcal{P}(N_i(w), P^2(w))$, $\mathcal{P}(w, N_i(P^2(w)))$, $\mathcal{P}(N_h(P^2(z)), z)$, and $\mathcal{P}(P^2(z), N_h(z))$ constructed by Algorithm Build $\mathcal{C}_3(s, t)$, $l(\mathcal{P}(w, N_i(P^2(w)))) = l(\mathcal{P}(N_h(P^2(z)), z)) = 4$ and $l(\mathcal{P}(N_i(w), P^2(w))) = 4$ (5) when $P(w) \neq P(N_i(w))$ ($P(w) = P(N_i(w))$), $l(\mathcal{P}(P^2(z), N_h(z))) = 4$ (5) when $P(z) \neq P(N_h(z))$ ($P(z) = P(N_h(z))$). In the following, Lemma 10 reveals that $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ form a $\mathcal{C}_3(s, t)$.

Lemma 10. $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ constructed by Algorithm Build $\mathcal{C}_3(s, t)$ form a $\mathcal{C}_3(s, t)$.

Proof. According to Lemma 8 and the structures of $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ as shown in Fig. 3(a), to prove $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ are disjoint to each other, two parts are necessary to be proved. First, $\mathcal{P}(w, N_i(P^2(w)))$ and $\mathcal{P}(N_i(w), P^2(w))$ are disjoint to each other, and $\mathcal{P}(w, N_i(P^2(w)))$ ($\mathcal{P}(N_i(w), P^2(w))$) is disjoint to $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ ($\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_3^3(s, t)$). Second, $\mathcal{P}(P^2(z), N_h(z))$ and $\mathcal{P}(N_h(P^2(z)), z)$ are disjoint to each other, and $\mathcal{P}(P^2(z), N_h(z))$ ($\mathcal{P}(N_h(P^2(z)), z)$) is disjoint to $\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ ($\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_2^3(s, t)$).

According to the structures of $\mathcal{P}(w, N_i(P^2(w)))$ and $\mathcal{P}(N_i(w), P^2(w))$ built by Algorithm Build $\mathcal{C}_3(s, t)$, all nodes of $\mathcal{P}(w, N_i(P^2(w)))$ ($\mathcal{P}(N_i(w), P^2(w))$) are not included in $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ ($\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$). Therefore, $\mathcal{P}(w, N_i(P^2(w)))$ ($\mathcal{P}(N_i(w), P^2(w))$) is disjoint to $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ ($\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$). Moreover, nodes of $\mathcal{P}(w, N_i(P^2(w)))$ are not included in $\mathcal{P}(N_i(w), P^2(w))$ as shown in Fig. 4. Therefore, $\mathcal{P}(w, N_i(P^2(w)))$ and $\mathcal{P}(N_i(w), P^2(w))$ are disjoint to each other.

Similarly, $\mathcal{P}(P^2(z), N_h(z))$ and $\mathcal{P}(N_h(P^2(z)), z)$ are disjoint to each other, and $\mathcal{P}(P^2(z), N_h(z))$ ($\mathcal{P}(N_h(P^2(z)), z)$) is disjoint to $\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_3^3(s, t)$.

$(\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_2^3(s, t)$).

Therefore, $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ constructed by Algorithm Build_ $\mathcal{C}_3(s, t)$ are disjoint to each other and form a $\mathcal{C}_3(s, t)$. ■

Therefore, according to the determined w and z , $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ can be built by Algorithm Build_ $\mathcal{C}_3(s, t)$ and form a $\mathcal{C}_3(s, t)$. Lemma 11 shows that length of $\mathcal{C}_3(s, t)$ is at most $\lceil (10n+2)/3 \rceil$ when it is built by Algorithm Build_ $\mathcal{C}_3(s, t)$.

Lemma 11. When $\mathcal{C}_3(s, t)$ is built by Algorithm Build_ $\mathcal{C}_3(s, t)$, $l(\mathcal{C}_3(s, t)) \leq \lceil (10n+2)/3 \rceil$ for $n \geq 4$.

Proof. Referring to Fig. 3(a), $l(\mathcal{P}_1^3(s, t)) = l(\mathcal{P}_2(s, t)) - l(\mathcal{P}^*(s, N_i(s))) - l(\mathcal{P}^\#(N_i(s), N_i(P^2(w)))) + d(s, w) + l(\mathcal{P}(w, N_i(P^2(w)))) = l(\mathcal{P}_2(s, t)) - l(\mathcal{P}^\#(N_i(s), N_i(P^2(w)))) + (l - l_w + 2) + 1$. Similarly, $l(\mathcal{P}_2^3(s, t)) = l(\mathcal{P}_3(s, t)) - l(\mathcal{P}^\#(N_h(P^2(z)), N_h(t))) + (m - l_z + 2) + 1$ and $l(\mathcal{P}_3^3(s, t)) = l(\mathcal{P}_1(s, t)) - (l - l_w + 2) + l(\mathcal{P}^\#(N_i(s), N_i(P^2(w)))) - (m - l_z + 2) + l(\mathcal{P}^\#(N_h(P^2(z)), N_h(t))) + l(\mathcal{P}(N_i(w), P^2(w))) + l(\mathcal{P}(P^2(z), N_h(z))) - l(\mathcal{P}^\#(N_i(w), N_i(P^2(w)))) - l(\mathcal{P}^\#(N_h(P^2(z)), N_h(z))) + 2$. As mentioned, there are $\lfloor (2 \times (l(\mathcal{P}^\#(N_i(s), N_i(s'))) - l(\mathcal{P}^*(s, s')) - 1)/3 \rfloor$ mesh edges in $\mathcal{P}^\#(N_i(s), N_i(P^2(w)))$ and $\lfloor (2 \times (l(\mathcal{P}^\#(N_h(t'), N_h(t))) - l(\mathcal{P}^*(t', t))) - 1)/3 \rfloor$ mesh edges in $\mathcal{P}^\#(N_h(P^2(z)), N_h(t))$. Referring to Fig. 4, $l(\mathcal{P}(w, N_i(P^2(w)))) = 4$; when $l(\mathcal{P}^\#(N_i(w), N_i(P^2(w)))) = 4$, $l(\mathcal{P}(N_i(w), P^2(w))) = 5$; when $l(\mathcal{P}^\#(N_i(w), N_i(P^2(w)))) = 3$, $l(\mathcal{P}(N_i(w), P^2(w))) = 4$. Hence, $l(\mathcal{P}_1^3(s, t)) \leq (4n-2) - \lfloor (2(n-2)-1)/3 \rfloor + 1 = \lceil (10n+2)/3 \rceil$. Both $l(\mathcal{P}_2^3(s, t))$ and $l(\mathcal{P}_3^3(s, t))$ can be calculated similarly, $l(\mathcal{P}_2^3(s, t)) \leq 2n+2\lfloor (2n+1)/3 \rfloor \leq \lceil (10n+2)/3 \rceil$ and $l(\mathcal{P}_3^3(s, t)) \leq \lceil (10n+2)/3 \rceil$. ■

According to Lemma 11, $l(\mathcal{C}_3(s, t)) \leq \lceil (10n+2)/3 \rceil$ when $\lceil (10n-4)/3 \rceil < l(\mathcal{P}_2(s, t)) \leq l(\mathcal{P}_3(s, t))$. Similarly, when $l(\mathcal{P}_2(s, t)) \leq \lceil (10n-4)/3 \rceil < l(\mathcal{P}_3(s, t))$, a $\mathcal{C}_3(s, t)$ can be formed as $\mathcal{P}_1^3(s, t)$, $\mathcal{P}_2^3(s, t) = \mathcal{P}_2(s, t)$ and $\mathcal{P}_3^3(s, t)$ as shown in Fig. 3(b), and $\mathcal{P}_1^3(s, t)$ and $\mathcal{P}_3^3(s, t)$ are not longer than $\lceil (10n+2)/3 \rceil$. Therefore, the following Corollary 12 can be obtained.

Corollary 12. $l(\mathcal{C}_3(s, t)) \leq \lceil (10n+2)/3 \rceil$, for $n \geq 4$.

Corollary 12 indicates an upper bound of $d_3(\text{PM}[n])$. In the following, a lower bound of ω -wide distance between a special pair of nodes is calculated, and then the lower bound of $d_\omega(\text{PM}[n])$ can be obtained.

3.3 ω -Wide Diameters

Let $s = (n; 0, 0)$ and $t = (n; 2^n-1, 2^n-1)$ are corners at layer n , so that $\forall u \in \mathcal{P}^*(s, t)$, u is a corner with two neighbors, and $l(\mathcal{P}^\#(N_i(u), N_i(P(u)))) = 2$. For ease of discussion, in the rest of this section, $\mathcal{C}_\omega(s, t)$ is a set of ω $\mathcal{P}(s, t)$ s, and all $\mathcal{P}(s, t)$ s are disjoint to a $\mathcal{P}^*(s, t) = \mathcal{P}^*(s, P^{n-l}(s)) \bullet \mathcal{P}^*(M_i; P^{n-l}(s), P^{n-l}(t)) \bullet \mathcal{P}^*(P^{n-l}(t), t)$ for $l=0$ or 1 , where s and $t \in V(\text{PM}[n])$. Before determining a lower bound of $d_\omega(s, t)$, some properties of paths

are shown in the following lemmas.

Lemma 13. $l(\mathcal{P}(s, t)) \geq 4n-2$.

Proof. Let a and $b \in V(\mathcal{P}(s, t))$, $l_a = l_b = l = \min\{l_v \mid v \in V(\mathcal{P}(s, t))\}$, be nodes in $\mathcal{P}(s, t)$ nearest to $P^{n-l}(s)$ and $P^{n-l}(t)$, respectively, and $\mathcal{P}(s, t) = \mathcal{P}(s, a) \bullet \mathcal{P}(M_i; a, b) \bullet \mathcal{P}(b, t)$. If a (b) is not a neighbor of $P^{n-l}(s)$ ($P^{n-l}(t)$), then apply Lemma 3 and Corollary 4 to $\mathcal{P}(s, a)$ ($\mathcal{P}(b, t)$) and thus a (b) can be replaced by $a' = N_i(P^{n-l}(s))$ ($b' = N_j(P^{n-l}(t))$) without increasing the length of $\mathcal{P}(s, t)$. Hence, $\mathcal{P}(s, t)$ can be rebuilt as $\mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), a') \bullet \mathcal{P}^*(M_i; a', b') \bullet \mathcal{P}^\#(b', N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$ without increasing its length and is still disjoint to $\mathcal{P}^*(s, t)$. As mentioned above, the rebuilt $\mathcal{P}(s, t)$ is the shortest one among all $\mathcal{P}(s, t)$ s with $\min\{l_v \mid v \in V(\mathcal{P}(s, t))\} = l$. In other words, under this condition, $l(\mathcal{P}(s, t))$ is at least $1+2(n-l)+2(2^l-1)-2+2(n-l)+1$. To minimize $l(\mathcal{P}(s, t))$, l should be 1 or 2 and thus $l(\mathcal{P}(s, t)) \geq 4n-2$. ■

The proof of Lemma 13 reveals that the layers of all nodes in every $\mathcal{P}(s, t)$ are at least 2 and any $\mathcal{P}(s, t)$ can be rebuilt as $\mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(P^{n-2}(s))) \bullet \mathcal{P}^*(M_2; N_i(P^{n-2}(s)), N_j(P^{n-2}(t))) \bullet \mathcal{P}^\#(N_j(P^{n-2}(t)), N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$ without increasing its length and is still disjoint to $\mathcal{P}^*(s, t)$.

Given a $\mathcal{C}_2(s, t)$ in which every path is disjoint to a $\mathcal{P}^*(s, t)$, according to the proof of Lemma 13, every $\mathcal{P}(s, t)$ s can be reconstructed as $\mathcal{P}^*(s, N_i(s)) \bullet \mathcal{P}^\#(N_i(s), N_i(P^{n-2}(s))) \bullet \mathcal{P}^*(M_2; N_i(P^{n-2}(s)), N_j(P^{n-2}(t))) \bullet \mathcal{P}^\#(N_j(P^{n-2}(t)), N_j(t)) \bullet \mathcal{P}^*(N_j(t), t)$. Since the degrees of both s and t are 3, a $\mathcal{C}_3(s, t)$ must be formed by add a $\mathcal{P}^*(s, t)$ to the built $\mathcal{C}_2(s, t)$. The $\mathcal{C}_2(s, t)$ should be transformed into $\mathcal{C}_2'(s, t)$ to shorten its total length with the aid of the proof of Lemma 13, and the $\mathcal{C}_3(s, t)$ is transformed into a $\mathcal{C}_3'(s, t)$. According to Lemma 13, $l(\mathcal{C}_3'(s, t)) \geq 4n-2$ and $l_T(\mathcal{C}_3'(s, t)) = l_T(\mathcal{C}_2'(s, t)) + d(s, t) \geq 2 \times (4n-2) + 2n = 10n-4$. Clearly, $\mathcal{C}_3'(s, t)$ is longer than the upper bound of 3-wide diameter shown in Corollary 12, thus $\mathcal{C}_3'(s, t)$ is not a $\mathcal{C}_3(s, t)$. To shorten length of $\mathcal{C}_3'(s, t)$, each path should share subpaths of $\mathcal{P}^*(s, t)$ during the $\mathcal{C}_3'(s, t)$ is transforming into a $\mathcal{C}_3''(s, t)$, such $l_T(\mathcal{C}_3''(s, t)) \leq l_T(\mathcal{C}_3'(s, t))$. In order to reduce total length of $\mathcal{C}_3''(s, t)$, during the sharing process, each path in $\mathcal{C}_3''(s, t)$ should include a subpath of $\mathcal{P}^*(s, t)$ from s and t , respectively. For ease of discussion, $\mathcal{C}_3''(s, t) = \{\mathcal{P}_1''(s, t), \mathcal{P}_2''(s, t), \mathcal{P}_3''(s, t)\}$. Without loss of generality, $\mathcal{P}_1''(s, t) = \mathcal{P}^*(s, a) \bullet \mathcal{P}(a, b) \bullet \mathcal{P}(b, t)$, $\mathcal{P}_2''(s, t) = \mathcal{P}(s, c) \bullet \mathcal{P}(c, d) \bullet \mathcal{P}^*(d, h) \bullet \mathcal{P}(h, g) \bullet \mathcal{P}(g, t)$ and $\mathcal{P}_3''(s, t) = \mathcal{P}(s, f) \bullet \mathcal{P}(f, e) \bullet \mathcal{P}^*(e, t)$ as shown in Fig. 5, where a and d (e and h) are nodes in $\mathcal{P}^*(s, P^{n-1}(s))$ ($\mathcal{P}^*(t, P^{n-1}(t))$), and paths $\mathcal{P}(a, b)$, $\mathcal{P}(c, d)$, $\mathcal{P}(f, e)$, and $\mathcal{P}(h, g)$ are built for sharing subpaths of $\mathcal{P}^*(s, t)$ to paths in $\mathcal{C}_3''(s, t)$. Naturally, all of them should disjoint to each other, and d is an ancestor of a , h is an

ancestor of e , then the formed $\mathcal{C}_3''(s, t)$ can be a container. In order to minimize lengths of $\mathcal{P}(b, t)$, $\mathcal{P}(s, f)$, $\mathcal{P}(s, c)$, and $\mathcal{P}(g, t)$, all of them are subpaths in $\mathcal{C}_3''(s, t)$.

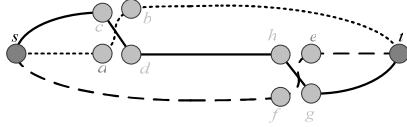


Fig. 5. The structures of $\mathcal{P}'_1(s, t)$ (dotted line), $\mathcal{P}'_2(s, t)$ (solid line) and $\mathcal{P}'_3(s, t)$ (dashed line).

Lemma 14. $l_T(\mathcal{C}_3''(s, t)) \geq 10n+4$, for $n \geq 4$.

Proof. In order to shorten total length of $\mathcal{C}_3''(s, t)$, $\mathcal{P}(a, b)$, $\mathcal{P}(f, e)$, $\mathcal{P}(c, d)$, and $\mathcal{P}(h, g)$ should be built as short as possible. Consider the structures of $\mathcal{P}(a, b)$ and $\mathcal{P}(c, d)$.

Case 1. (b is $N(a)$): Clearly, c must be a descendant of a and $\mathcal{P}(c, d)$ must be built as $\mathcal{P}^*(M_{ic}; c, u) \bullet \mathcal{P}^*(u, d)$ and still disjoint to $\mathcal{P}'_1(s, t)$ and $\mathcal{P}'_3(s, t)$, where u is the descendant of d at layer l_c nearest to c . That is, the two neighbors of a are not included in $\mathcal{P}(c, d)$ which are included in $\mathcal{P}'_2(s, t)$ and $\mathcal{P}'_3(s, t)$ respectively. Without loss of generality, $N_0(a)$ is included in $\mathcal{P}'_2(s, t)$ and $N_1(a)$ is included in $\mathcal{P}'_3(s, t)$. In order to shorten $l(\mathcal{P}'_2(s, t))$, $d = P(a)$ and c is $N_0(v)$ according to Lemma 3. Obviously, before $\mathcal{P}(c, d)$ is built, the total length is $l_T(\mathcal{C}_3''(s, t)) - 2$.

Case 1.1. ($\mathcal{P}'_3(s, t)$ is rebuilt as $\mathcal{P}_3''(s, t)$): Let $\mathcal{P}(N_1(v), N_1(d)) = \mathcal{P}^*(M_{ic}; N_1(v), w) \bullet \mathcal{P}^*(w, N_1(d))$ is a subpath of $\mathcal{P}_3''(s, t)$, where v is the diagonal node of c in a $PM[1]$, and w is $N_1(N_1(N_1(v)))$. Then $l(\mathcal{P}_3''(s, t)) \geq l(\mathcal{P}'_3(s, t)) + 1$ and $l(\mathcal{P}(c, d))$ is at least 4. That is, $\mathcal{P}(N_1(v), N_1(d))$ and $\mathcal{P}(c, d)$ will contribute total length at least 5. Thus, sharing a subpath of $\mathcal{P}^*(s, t)$ costs at least 3. Hence, $l_T(\mathcal{C}_3''(s, t)) \geq (10n-2) + (2 \times 3) = 10n+4$.

Case 1.2. (Otherwise): $u = N_0(N_1(N_1(c)))$ and $l(\mathcal{P}(c, d))$ is at least 5. Thus, sharing a subpath of $\mathcal{P}^*(s, t)$ costs at least 3. Hence, $l_T(\mathcal{C}_3''(s, t)) \geq (10n-2) + (2 \times 3) = 10n+4$.

Case 2. (b is $C(a)$): Clearly, c is a $C^2(a)$.

Case 2.1. ($\mathcal{P}'_3(s, t)$ is rebuilt as $\mathcal{P}_3''(s, t)$): Let $\mathcal{P}(N_1(v), N_1(d)) = \mathcal{P}^*(M_{ic}; N_1(v), w) \bullet \mathcal{P}^*(w, N_1(d))$ is a subpath of $\mathcal{P}_3''(s, t)$, where v is the diagonal node of c in $PM[1]$, and w is $N_1(N_1(N_1(v)))$. Then $l(\mathcal{P}_3''(s, t)) \geq l(\mathcal{P}'_3(s, t)) + 1$ and $l(\mathcal{P}(c, d))$ is at least 6. That is, $\mathcal{P}(N_1(v), N_1(d))$ and $\mathcal{P}(c, d)$ will contribute total length at least 7. Thus, sharing a subpath of $\mathcal{P}^*(s, t)$ costs at least 5. Hence, $l_T(\mathcal{C}_3''(s, t)) \geq (10n-2) + (2 \times 5) = 10n+8$.

Case 2.2. (Otherwise): and u a $C^2(N_0(N_1(a)))$, thus $d(M_{ic}; c, u)$ is at least $2 \times (2^2 - 1) = 6$ and length of $\mathcal{P}(c, d)$ is at least 8. Obviously, before $\mathcal{P}(c, d)$ is built, the total length is

$l_T(\mathcal{C}_3''(s, t)) - 2$. Thus, sharing a subpath of $\mathcal{P}^*(s, t)$ costs at least 6. Hence, $l_T(\mathcal{C}_3''(s, t)) \geq (10n-2) + (2 \times 6) = 10n+10$.

Therefore, the total length of $\mathcal{C}_3''(s, t)$ is at least $10n+4$. ■

According to Lemma 14, the following corollary can be obtained.

Corollary 15. $d_3(s, t) \geq \lceil (10n+2)/3 \rceil$, for $n \geq 4$.

After both upper and lower bounds are determined in Corollary 12 and Corollary 15, 3-wide diameter of $PM[n]$ are shown in the following theorem.

Theorem 16. $d_3(PM[n]) = \lceil (10n+2)/3 \rceil$, for $n \geq 4$.

It is easy to check that $d_3(PM[1]) = 3$, $d_3(PM[2]) = 6$ and $d_3(PM[3]) = 10$.

4 Concluding Remarks

Parallelism, transmission delay and fault-tolerance are very important issues for parallel communication on an interconnection network. The ω -wide diameter of an interconnection network is a metric of evaluating parallel communication performance and reliability of the network. This work computes $d_\omega(PM[n])$ for $\omega=3$. It is found that $d_3(PM[n]) = \lceil (10n+2)/3 \rceil$ for $n \geq 4$. Furthermore, $d_3(PM[1]) = 3$, $d_3(PM[2]) = 6$, and $d_3(PM[3]) = 10$.

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